

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013–14

Paper: CSB-602 — Data and File Structures

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer **five** questions in all, including Question 1 which is compulsory. Figures on the right-hand side indicate marks.*

Question 1. Answer **all** parts:

[3 × 6 = 18]

- (a) Define the terms: Abstract Data Types (ADT) and Data Structure.
- (b) State the advantages of using postfix notation over infix notation.
- (c) Define Double-ended Queue (Deque).
- (d) What is a Binary Heap? Distinguish between Max-Heap and Min-Heap.
- (e) List the basic operations carried out on a linked list.
- (f) Briefly discuss the following terms: Tree, Degree of a node, and Height of a tree.
- (g) What do you mean by collision in hashing? Briefly explain any two collision resolution methods.

Question 2.

[2 + 5 + 6 = 13]

- (a) Define ADT Array.
- (b) What is a sparse matrix? Give a structure for its efficient representation.
- (c) Give an algorithm for addition of two sparse matrices stored in the representation you proposed above.

Question 3.

[2 + 6 + 5 = 13]

- (a) What are the two important criteria to be satisfied by any recursive function?
- (b) Explain the implementation of a stack using an array with algorithms for all operations on a stack.
- (c) Write an algorithm to convert an infix expression to a postfix expression, and explain it through an example.

Question 4.

[3 + 10 = 13]

- (a) What do you mean by order of growth in reference to time complexity of an algorithm?
- (b) Explain the quick sort algorithm with an example. Also mention any three strategies for choosing a pivot element.

Question 5.

[4 + 9 = 13]

- (a) Define full binary tree and complete binary tree with an example for each.
- (b) Explain binary tree traversals (in-order, pre-order, post-order) with recursive algorithms.

Question 6.

[7 + 6 = 13]

- (a) Explain Breadth First Search (BFS) traversal on a graph with an example.
- (b) Write an algorithm for adding two polynomials using a linked list.

Question 7. Write notes on any **two** of the following:

[6.5 × 2 = 13]

- (a) Hashing
- (b) Binary Search Algorithm (Iterative and Recursive)
- (c) Sequential and Index-Sequential file organization techniques
- (d) External sorting methods with an example